

Reviewed Mathematical Methodology for the RAI Risk Taxonomy Space

City-scale AI Risk Co-occurrence and Socio-technical Cascading Propagation

RAI Risk Taxonomy 2.0 Methodology Note

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Abstract

This note defines a reviewed methodology for connecting the RAI Risk Taxonomy to city-scale AI research. The primary object is not a two-dimensional embedding map. It is a typed, multilayer evidence network linking L4 risks, human and institutional actors, technical mediators, data objects, urban contexts, outcomes, controls, and observation units. Co-occurrence represents conditional association. Directional propagation is reported only when temporal order and a specific transmission or transformation mechanism are supported. The term contagion is restricted to calibrated dynamic models and is not used as a synonym for co-occurrence. The method is designed as an Urban AI Risk Observatory that can support collaboration between responsible AI evaluation and city-scale sensing, mobility, public-space, and environmental research.

1 Purpose and Research Position

The RAI Risk Taxonomy contains 1,711 registered L4 identifiers, including 1,660 active cards and 51 merged records retained for traceability. The proposed Space connects the active taxonomy to urban AI deployments without changing the released L0 to L4 hierarchy.

Senseable City Seoul studies social interaction in urban space using visual AI, big-data analytics, and sensing platforms. Its stated research scope includes variation across age, socioeconomic background, time, season, and climate [13]. This creates a suitable empirical setting for studying how AI risks combine across data collection, model operation, institutional decisions, public services, physical infrastructure, and community outcomes.

The recommended platform name is:

RAI Risk Taxonomy Space: Urban AI Risk Observatory

The recommended research programme is:

City-scale AI Risk Co-occurrence and Socio-technical Cascading Propagation

This terminology separates three concepts that must not be conflated:

- co-occurrence, which is an observed association within a defined urban observation;
- propagation mechanism, which is a plausible or evidenced path through data, technology, decisions, and actors;
- causal transition, which requires temporal order and evidence that rules out credible alternative explanations.

2 Independent Methodological Review

Two independent expert-agent reviews were conducted before revision. One review focused on complex networks, multilayer systems, and cascading risk. The other focused on urban AI, sociotechnical governance, and living-lab research design. Both reviews reached the same central conclusion: the research direction is academically viable, but co-occurrence must not be described as contagion or causal propagation without additional evidence.

Table 1: Major review findings and adopted revisions

Issue	Review finding	Adopted revision
Risk contagion	Risks transform rather than reproduce unchanged	Use cascading propagation as the general term; reserve contagion for calibrated dynamic models
Actor definition	Human, institutional, and technical objects were mixed	Separate actors, technical mediators, data objects, contexts, outcomes, and controls
Co-occurrence	Joint observation does not establish direction or cause	Introduce three explicit evidence levels
Network form	Pairwise projection loses event-level context	Preserve the typed hypergraph and use projections only for exploration
L3 and HOLD	Attractor and boundary language was stronger than the evidence	Treat L3 as a taxonomy family and HOLD as a review status
Centrality	Structural position can be mistaken for risk magnitude	Prohibit interpretation of centrality as severity or probability
Validation	Semantic similarity alone is insufficient	Add coding, null-model, bootstrap, temporal, fairness, and cross-city validation

3 Conceptual Ontology

3.1 Typed Objects

Let the city-scale risk system be

$$\mathcal{S} = (\mathcal{R}, \mathcal{A}, \mathcal{T}, \mathcal{D}, \mathcal{U}, \mathcal{O}, \mathcal{K}, \mathcal{Q}), \quad (1)$$

where:

The term actor is restricted to persons and organisations capable of action, decision, authority, or responsibility. A dataset, model, or sensor is a technical mediator or data object unless Actor Network Theory is explicitly adopted. This distinction preserves accountability and avoids assigning normative agency to technical components.

3.2 L4 Logical Roles

L4 cards may describe different stages of a risk pathway. Each active card should therefore receive one or more logical-role attributes:

$$\text{hazard} \rightarrow \text{failure mode} \rightarrow \text{exposure} \rightarrow \text{outcome}, \quad (2)$$

Table 2: Core object types

Set	Definition	Examples
$\mathcal{R}$	Active L4 risk cards	privacy loss, sensor failure, allocation bias
$\mathcal{A}$	Human and institutional actors	citizens, operators, city agencies, suppliers
$\mathcal{T}$	Technical mediators	models, sensors, platforms, robots, APIs
$\mathcal{D}$	Data objects	urban imagery, mobility traces, environmental streams
$\mathcal{U}$	Urban contexts	public spaces, transport networks, neighbourhoods
$\mathcal{O}$	Outcomes	exclusion, physical harm, trust erosion
$\mathcal{K}$	Controls	access control, human review, audit records, fallback procedures
$\mathcal{Q}$	Observation units	deployment episodes, incidents, bounded scenarios

with control failure represented as a cross-cutting condition. This role annotation prevents a failure mechanism and its downstream harm from being interpreted as equivalent nodes merely because they occur together.

3.3 Observation Unit

The observation unit is more specific than a full report or project:

$$q = (p, z, t, g, \tau, \ell), \quad (3)$$

where p is an AI deployment or use case, z is an urban spatial unit, t is a time interval, g is an affected group, τ is an AI task, and ℓ is a lifecycle stage. A paper, project page, or incident report is an evidence source supporting one or more observations, not the observation itself.

This design limits bias from document length, broad project descriptions, and repeated terminology. It is consistent with lifecycle and contextual approaches in the NIST AI Risk Management Framework and the OECD classification framework [9, 10].

4 Higher-order Network Representation

4.1 Typed Hypergraph

The primary representation is a typed hypergraph

$$\mathcal{H} = (V, \mathcal{E}, \psi, \rho), \quad (4)$$

where V is the union of all typed objects, $\mathcal{E}$ is the set of hyperedges, $\psi(v)$ returns the type of node v , and $\rho(e)$ records the evidence level of hyperedge e .

Each observation q produces a hyperedge

$$e_q = \{r_i, a_j, t_k, d_m, u_n, o_s, k_v\}, \quad (5)$$

containing the risks and sociotechnical elements jointly involved in that bounded urban episode. Hyperedges preserve group interactions that would be lost when every observation is decomposed into pairs

[2].

4.2 Multilayer Views

For interpretation, the hypergraph can be viewed as a multilayer network:

$$\mathcal{L} = \{\text{data, technology, decision, institution, space, outcome, control}\}. \quad (6)$$

Layer-specific relationships are retained separately. A combined score may be calculated for a defined analysis, but the public interface must preserve the relationship type. Multilayer representation is necessary because connectivity in one layer can produce different behaviour from connectivity in another [6, 3].

5 Co-occurrence Estimation

5.1 Context to Risk Incidence

Let

$$X_{qi} \in [0, 1] \quad (7)$$

denote the coding confidence that L4 risk r_i is present in observation q . Binary coding is obtained when $X_{qi} \in \{0, 1\}$.

The raw co-occurrence count is

$$C_{ij} = \sum_{q \in \mathcal{Q}} X_{qi} X_{qj}. \quad (8)$$

Raw counts are not sufficient because frequent risks and large observation units can dominate the projection.

5.2 Normalized Association

For binary observations, the Jaccard association is

$$J_{ij} = \frac{C_{ij}}{n_i + n_j - C_{ij}}, \quad n_i = \sum_q X_{qi}. \quad (9)$$

Normalized pointwise mutual information is

$$\text{nPMI}_{ij} = \frac{\log(p_{ij}/p_i p_j)}{-\log p_{ij}}, \quad (10)$$

where p_i , p_j , and p_{ij} are estimated from the defined observation universe. Jaccard measures proportional overlap; nPMI measures departure from independent occurrence.

5.3 Statistically Validated Projection

Each candidate risk to risk edge is tested against a bipartite null model that preserves risk prevalence and observation size. Edge-specific p -values are corrected with the Benjamini and Hochberg false discovery rate procedure. Only edges satisfying all of the following should appear in the default public network:

- a minimum observation-support threshold;
- corrected statistical significance;

- a minimum normalized-association threshold;
- stable inclusion under observation-level bootstrap resampling.

This follows statistically validated bipartite-network projection rather than unfiltered one-mode projection [14, 4].

6 Evidence-graded Propagation

6.1 Three Evidence Levels

Table 3: Relationship evidence levels

Level	Required evidence	Permitted interpretation
E1 Association	Joint presence in a bounded observation with validated association	Risks co-occur more often than expected
E2 Mechanism	E1 plus a documented data, technical, decision, or actor-mediated pathway	A plausible cascading pathway is supported
E3 Temporal	E2 plus repeated temporal order and validation against alternatives	Directional propagation is empirically supported

The following implication is invalid:

$$C_{ij} > 0 \not\Rightarrow r_i \rightarrow r_j. \quad (11)$$

A directional relationship requires temporal precedence, an explicit mechanism, and evidence that addresses common causes and selection effects. Causal language additionally requires an intervention, quasi-experiment, or a defensible causal-identification design.

6.2 Dynamic Model

When event time is available and observations are sufficiently dense, a marked multivariate Hawkes process may represent mutually exciting risk events:

$$\lambda_j(t) = \mu_j + \sum_i \int_0^t \phi_{ij}(t-s) dN_i(s). \quad (12)$$

Here, μ_j is the baseline event intensity and ϕ_{ij} is a time-dependent excitation kernel. Marks encode mediators, actor types, locations, affected groups, and controls [5].

When temporal observations are sparse, the method must use a scenario-based stochastic cascade instead of estimating transmission probabilities. Such output is labelled a counterfactual scenario, not a city-scale forecast.

SIR is not the default model. Urban AI risks can coexist, recur, persist in stored data, transform across lifecycle stages, and be reduced by controls. These properties violate the simplest assumptions of pathogen transmission models [11, 16].

7 Network Measures and Interpretation Rules

The following measures support diagnosis:

- weighted degree, which describes the breadth of validated association;
- betweenness, which identifies entities lying on many network paths;
- participation coefficient, which identifies entities connecting multiple layers or communities;
- community structure, which describes mesoscopic groups of co-occurring risks;
- cascade size under a specified scenario, which estimates potential downstream activation under explicit assumptions.

Interpretation constraints are mandatory:

- centrality is not severity, probability, or causal influence;
- network-layout distance is not semantic similarity or causal distance;
- L3 is a released taxonomy family used for colour and filtering, not a dynamical attractor;
- HOLD is a taxonomy-review status, not a structural boundary unless network evidence independently supports that interpretation;
- Physical AI is not called a stable subsystem without direct structural and temporal evidence.

8 Validation Protocol

8.1 Content and Coding Validity

Urban researchers, AI safety researchers, public-sector practitioners, and affected-community representatives review the ontology and pathway templates. Two independent coders annotate a shared sample. Reliability is reported with Krippendorff's alpha for categorical annotations and class-specific precision, recall, and F1 for multilabel extraction.

8.2 Network Statistical Validity

- compare observed edges with degree-preserving bipartite null models;
- report corrected significance, association size, and support count;
- calculate edge-inclusion probability through project or observation bootstrap;
- test sensitivity to minimum support, Jaccard, nPMI, and ontology granularity;
- repeat analyses after removing the largest projects and most frequent risks.

8.3 Structural Stability

Community assignments are compared using adjusted Rand index or variation of information. Centrality rank stability is evaluated across bootstrap replicates. Layer-specific results are compared with combined-network results to identify artefacts introduced by aggregation.

8.4 Temporal and External Validation

When temporal data exist, an earlier interval is used for estimation and a later interval for evaluation. Recommended measures include log loss, Brier score, area under the precision-recall curve, and calibration.

The dynamic model is compared with an independent-occurrence baseline and a static co-occurrence baseline.

External validation uses leave-one-city-out or leave-one-project-type-out designs. Seoul can serve as the initial study site, while other urban contexts test transferability rather than being assumed equivalent.

8.5 Fairness, Privacy, and Participation

Validation is stratified by age, disability, gender where lawfully and ethically available, socioeconomic context, and geographic area. The study compares detection rates, false negatives, and uncertainty across groups. Urban imagery and sensor data require purpose limitation, minimum necessary collection, bounded retention, re-identification assessment, and limits on public spatial resolution. Living-lab participation must not treat routine presence in public space as research consent [15, 8].

9 Interface Specification

The public Space uses a Risk to Entity to Urban Context structure rather than a dimensionality-reduction scatter plot.

Table 4: Visual encoding rules

Visual element	Meaning
Risk node	Active L4 card; L3 appears as taxonomy colour or filter
Actor node	Human or institutional decision and accountability subject
Mediator node	Data object or technical component that transmits or transforms risk
Context node	Urban place, time, group, task, and lifecycle condition
Solid undirected edge	E1 statistically validated association
Solid directional edge	E2 mechanism-supported cascading pathway
Dotted directional edge	hypothesis or counterfactual simulation
Evidence label	E1 association, E2 mechanism, or E3 temporal validation
HOLD badge	taxonomy review state; not an independent risk family

Force-directed placement may improve readability, but geometry has no substantive meaning. Selecting an L4 card reveals the observation support, shared mediator, downstream decision or outcome, relationship evidence level, and uncertainty.

10 Joint Research Design

The staged empirical sequence is:

1. retrospective analysis of literature, incidents, and existing urban projects;
2. synthetic-data and digital-twin scenario analysis;
3. non-interventional field observation;
4. citizen and stakeholder review;
5. limited evaluation of risk controls;
6. independent assessment and public reporting of non-sensitive findings.

Table 5: Proposed KT RAI and MIT Senseable City Lab work packages

Package	KT RAI contribution	MIT SCL contribution	Joint output
WP1	L4 definitions and risk roles	Urban entity and context ontology	Urban AI Risk Ontology
WP2	Evidence grades and taxonomy review	Project and observation structure	Project to Risk to Entity registry
WP3	Co-occurrence and HOLD diagnosis	Spatial, temporal, and interaction layers	Multilayer risk network
WP4	Controls and residual risk	Urban cascade and feedback scenarios	Cascading-risk simulator
WP5	TEVV and independent review	Field protocol and civic participation	Seoul pilot protocol
WP6	Taxonomy reproducibility	Cross-city comparison	Transferability assessment

11 Conclusion

The reviewed methodology replaces a primary dimension-reduction view with an evidence-graded, typed network of city-scale AI risks. Its descriptive layer identifies statistically validated co-occurrence. Its mechanistic layer represents plausible transformations through data, models, services, decisions, actors, and controls. Its dynamic layer is activated only when temporal evidence supports propagation analysis.

The resulting Space is not a map of causal certainty. It is a structured Urban AI Risk Observatory that makes associations, mechanisms, uncertainty, and review status visible while preserving the RAI Risk Taxonomy as the reference classification.

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